How Much Fault Tolerance Is in a Fly Brain?

Fault-Injection Analysis of a Whole-Connectome Spiking Controller for Physical Intelligence

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***Abstract*—Whole-brain point-neuron models built from complete synaptic wiring diagrams have begun to serve as controllers in embodied settings, but their behavior under component-level faults has, to our knowledge, never been measured. We take the released leaky integrate-and-fire (LIF) whole-brain model of *Drosophila melanogaster* (138,639 neurons, 15.1M signed synapses) and inject five hardware-motivated fault classes—weight variability, synapse loss, random unit loss, hub loss, and sensory-channel deafferentation—scoring behavioral outcome on descending command neurons. Across 36 fault conditions (12 seeded trials each), the connectome absorbs 50% lognormal weight variability, retains 76% of escape-command activity after 50% synapse loss, and is immune to 10% random unit loss, while loss of the top-1% out-degree hubs produces a 2.5× runaway of global activity rather than silence. An edge-permuted Erdős–Rényi null network with identical weight and out-degree statistics transmits nothing, and an analytic multi-hop pathway analysis shows that relative pathway decay under edge loss is topology-independent: functional robustness arises from threshold amplification acting on convergence structure, not from graph redundancy alone. We distill the results into five design principles for fault-tolerant neuromorphic controllers.**

***Index Terms*—neuromorphic computing, connectome, fault tolerance, fault injection, Drosophila**

I. Introduction

PHYSICAL intelligence—agents whose perception, decision, and action loops close in the physical world under tight energy and latency budgets—demands controllers that remain robust on imperfect hardware [1], [2]. Neuromorphic systems promise the required efficiency [3], [4], and spiking implementations on Loihi 2 and FPGA substrates now cover vision and control workloads [5]–[7]. What remains poorly quantified is how *wiring* determines robustness: engineered spiking networks are designed task-by-task, so their tolerance to weight noise, synapse loss, and unit death must be re-measured per design [8], [9]. Complete, signed wiring diagrams of whole brains are now public, and point-neuron models built on them reproduce realistic dynamics without training [10]; an evolved connectome is thus a fully specified, field-tested controller and a reference point for benchmarking the fault tolerance of engineered designs.

We perform, to our knowledge, the first systematic *fault-injection* study on such a controller: the released leaky integrate-and-fire (LIF) whole-brain model of *Drosophila melanogaster*[1], which instantiates the complete FlyWire connectome [10]. We instrument a behavioral task—the visuomotor escape pathway, in which looming-sensitive visual interneurons (LC4, LPLC2) converge onto descending command neurons (DNp01, DNa02, DNp04, DNb01) that trigger take-off [11], [12]—and score outcome as command-population firing rate. The pathway is a complete sensor-to-command controller whose failure modes map directly onto hardware fault classes: weight imprecision, broken interconnect, dead cores, and dead sensors.

We inject five fault families (Section III-C), each over 12 seeded 600 ms trials, and separate wiring specificity from generic topology with two controls: an edge-permuted Erdős–Rényi (ER) null network with identical weight and out-degree statistics, and an analytic multi-hop pathway-mass computation. Our contributions: (i) a fault-injection framework mapping hardware failure modes onto a biologically exact 138k-neuron controller; (ii) the first behavioral-grade tolerance curves of a complete connectome (50% weight noise absorbed, 50% synapse loss retains 76%, 10% unit loss invisible); top-1% hub loss instead produces a 2.5× *activity runaway*, and whole-channel deafferentation is the only catastrophic mode; (iii) a decomposition showing that linear graph redundancy explains neither specificity nor graceful degradation, implicating threshold amplification in convergent spiking dynamics; and (iv) five design principles for fault-tolerant neuromorphic controllers.

II. Related Work

The release of complete wiring diagrams for the adult *Drosophila* brain [10] has shifted connectome research from static graph analysis to executable models spanning visual system convolutions [13], neuron-level response prediction [14], learned tasks and whole-body locomotion on whole-CNS wiring [15]–[17], and biofidelic processing units [18], with cell-type studies resolving descending-neuron input sharing [19], [20]. Closest to our substrate, the released LIF whole-brain model (footnote 1) reproduces global activity and has been ported to Loihi 2 [5]. None of these works varies the integrity of the controller; we treat the connectome controller as a fixed plant and sweep internal fault modes.

Robustness of spiking networks to weight noise has mostly been characterized on small trained architectures [8], [9]; purpose-built fault-injection frameworks for trained SNNs—SpikeFI [21], SpikingJET [22]—automate this but score task accuracy on human-designed networks, a line continued by STDP fault campaigns and population-coded redundancy [23],

[1]https://github.com/philshiu/Drosophila_brain_model

[24]. Hardware platforms expose complementary constraints: on-chip variability co-design [6], FPGA precision–capacity trade-offs [7], permanent interconnect faults [25], and edge energy–robustness envelopes [3], [26]; trained-CNN dependability analysis [27] completes this picture. Benchmarking efforts omit fault-injection axes [28], and formalizations of neuromorphic control do not yet treat degradation [29]. Insect-inspired collision-aversion circuits are robust by construction, but small and hand-designed [11], [12], [30]. What a complete connectome uniquely enables is tolerance measurement at the scale of an entire evolved controller; the resulting curves read as requirements rather than observations.

III. Fault-Injection Framework

A. Whole-Brain Spiking Substrate

We adopt the released whole-brain LIF model of *Drosophila* (Section I), implemented in Brian2 and built on the FlyWire connectome [10]. Every neuron is a current-based LIF unit (threshold -45 mV, reset -52 mV, τ_m=5 ms, refractory 2.2 ms, delay 1.8 ms). Every chemical synapse is a delta synapse of magnitude 0.275 mV signed by the presynaptic cell's neurotransmitter; the connectivity table yields 15,091,983 edges between 138,639 neurons, 40% of them inhibitory. Global activity is sustained by the released model's calibrated Poisson background drive. Our engine restores the full network state at the start of every trial, so all conditions share identical initial states and background realizations; one 600 ms trial runs in $\approx$3.5 s on one CPU core.

B. Behavioral Task: Visuomotor Escape

The instrumented pathway (Fig. 1) is the fly's looming-triggered escape circuit: looming-sensitive visual projection neurons LC4 (104) and LPLC2 (210) converge on descending neurons [11], [12]. Because the looming response of these cells contains a sustained high-rate component, we drive them—and, as a control, an equal-sized random neuron set—with independent 150 Hz Poisson trains through a stimulus layer gated per trial, rather than modeling looming kinetics. The behavioral readout is the population rate of the command set {DNp01, DNa02, DNp04, DNb01} (8 neurons) over the last 500 ms; all 1,299 descending neurons serve as a secondary readout. Calibration drove the command population to 94.6 Hz (interdecile 93.6–95.8 Hz, first-spike latency 4.4 ms on the 10-trial set) under escape stimulation but only 8.2 Hz (6.4–9.2 Hz) under random stimulation; we set $\theta = 51.4$ Hz midway between them; every pass/fail classification below is unchanged for any threshold between the distributions' observed trial extremes, 9.2 and 63.1 Hz. Calibration statistics refer to a 10-trial set; Section IV baselines use the 12 sweep trials.

C. Fault Taxonomy

Five fault families cover the hardware-relevant failure modes. F1 multiplies every synaptic weight by an i.i.d. lognormal factor ($\sigma \in \{0.05, 0.1, 0.2, 0.5\}$), modeling analog weight variability and bit-level imprecision. F2 deletes a random fraction $f \in \{0.05, 0.10, 0.25, 0.50\}$ of synapses, modeling

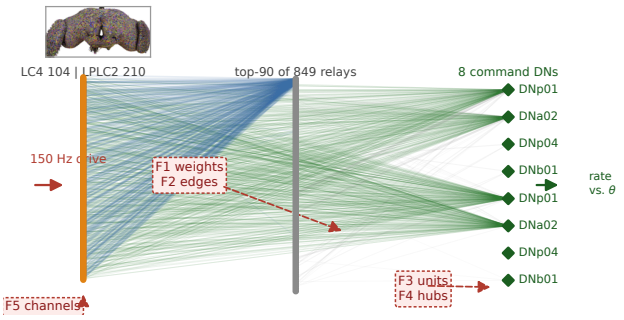


Fig. 1. The instrumented escape pathway, drawn from the connectome itself. Left: the 314 driven sensory cells (LC4, LPLC2); center: top-90 of the 849 two-hop relays; right: the 8 command DNs. Line width and opacity scale with synapse count (594 direct sensory→DN edges); dashed red marks the fault-injection points (F5 channels; F1/F2 synapses; F3/F4 units). Inset: reconstructed adult brain, adapted from Dorkenwald et al. [31] (CC BY 4.0).

interconnect or memristor failures. F3 silences all output synapses of a random fraction $f \in \{0.01, 0.02, 0.05, 0.10\}$ of units, excluding input and readout cells (equivalent to unit removal in this delta-synapse model), modeling dead cores. F4 applies the same operation to the top 1%, 2%, or 5% of units ranked by total outgoing synapse count (again excluding input and readout), modeling failure of high-fanout components; these cells carry 9–22% of all synapses (for the 1%, 2% and 5% sets, respectively), so F4 is a targeted, correlated version of F3. F5 zeroes all outgoing synapses of one (LC4 or LPLC2) or both sensory channels, modeling sensor or afferent-path failure. All faults are applied after state restoration and held fixed across the 12 trials of a condition, with SHA-256–derived deterministic realizations.[2]

D. Controls

Two controls isolate what the biological wiring contributes. The first is an ER null: permuting the postsynaptic endpoint of every synapse preserves out-degree, weight, and sign but destroys in-degree structure and convergence motifs; the identical sweep runs on it. The second is analytic: for both graphs we compute the *pathway mass* $m_k = \|A^k\|_1$ on the input-to-command submatrix (A holds synapse counts, i.e., absolute values of the signed connection strengths), the synapse-weighted mass of k-hop directed routes from LC4∪LPLC2 into the descending set, on the intact graph and after the F2 edge-deletion levels; comparing its decay with the simulated behavioral decay separates topological redundancy from dynamical amplification.

IV. Results

A. Baseline: The Connectome Is a Specific, Low-Latency Controller

Driving LC4 and LPLC2 at 150 Hz produces a command-population rate of 94.7 Hz (median over 12 trials; interdecile 94.0–95.6 Hz) with 4.05 ms median first-spike latency, far

[2]Code, connectome-derived edge lists, and per-trial records are available from the author.

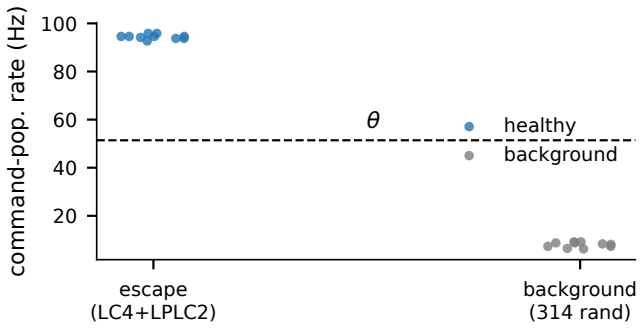


Fig. 2. Readout specificity. Command-population rate under escape stimulation (LC4+LPLC2) versus matched random stimulation; dashed line: θ.

above θ (Fig. 2). An equal-sized random set, driven identically, yields only 8.2 Hz on the same readout while producing *more* total activity (293k vs. 40k spikes): the readout is pathway-specific, not generic arousal. When both sensory channels are silenced ($F5_{\text{both}}$), the command rate collapses to exactly zero in every trial—consistent with effective drive passing only through the two lesioned channels—while the rest of the brain keeps spiking at background level: the score measures signal transmission through the instrumented pathway.

B. Tolerance Curves: Noise Is Absorbed, Disconnection Degrades Gracefully

Figure 3 shows dose–response curves for the two most consequential families; the F1 and F3 responses are flat to within noise (Table I). Weight variability (F1) is invisible: at σ=0.5 the rate is 94.7 Hz (healthy: 94.7 Hz), total activity up only from 39.7k to 41.5k spikes. Random synapse deletion (F2) degrades gracefully: 5% and 10% loss leave the rate at 91.9 and 93.1 Hz, 25% loss retains 84.7 Hz (89.5% of baseline), and even 50% loss retains 72.4 Hz (76.5%), with all 12 trials successful in every condition (the weakest per-trial rates, 63.1 Hz under single-channel loss and 68.1 Hz at 50% synapse loss, both remain far above θ). Disconnection's cost appears almost entirely in latency: 4.05 to 5.4 ms at 50% loss (+33%), consistent with recruitment of longer residual multi-hop routes. Random unit loss (F3) up to 10%—nearly 14,000 neurons with all their synapses—has no measurable effect (94.6 vs. 94.7 Hz): command drive integrates over hundreds of converging presynaptic populations, and no single 10% slice is load-bearing. The wide 1,299-neuron descending layer tracks the same pattern (2,941 spikes healthy vs. 1,918 at 50% synapse loss), so the graded response is a property of command encoding, not of the eight-neuron readout.

C. Hub Loss Produces Runaway, Not Silence

The dose–response of hub removal (F4) is non-monotonic and qualitatively different from every distributed fault (Figs. 3b, 4b). Removing the top 1% of cells ranked by total outgoing synapse count (1,386 neurons, 9.1% of all synapses) *increases* the command rate to 122.3 Hz (+29%, interdecile 121.1–123.9 Hz across trials) and more than doubles global activity (99.9k spikes, 2.52× healthy); at 2% the rate stays elevated (117.5 Hz) while activity partially normalizes (42.4k); at 5% (22.4% of synapses) both metrics return to near-baseline (94.1 Hz, 36.4k). Hub lesions do not starve the pathway of drive—they remove broadly connected inhibitory regulation, and spared circuitry over-drives the command layer: about 60% of removed hub output synapses are inhibitory, versus 40% network-wide. In an engineered system this is the dangerous mode: a controller that violates its output envelope while every per-unit health monitor still reads "active." The same lesion in the ER null produces no such signature, tying the runaway to biological hub structure rather than to degree heterogeneity per se.

D. What the Wiring Contributes: Null Network and Pathway Analysis

The ER control dissociates specificity from redundancy: with out-degree and weight preserved but in-degree structure permuted, the null transmits *nothing* to the command layer in every condition—0.0 Hz even on the 1,299-neuron broad readout—while global activity is normal. Pathway mass quantifies the convergence advantage: the intact connectome carries 5.8× more 1-hop and 17.7× more 4-hop route mass from sensory input to command neurons than the ER null. Yet the *relative* decay of pathway mass under the same edge-deletion levels is nearly identical in both graphs (e.g., at f=0.5, k=8: 0.3% vs. 0.4% of baseline; Fig. 4b). Combined with the simulation result that behavior degrades far more slowly than pathway mass (76% functional retention at f=0.5, against 5.8% retention of the intact four-hop route mass), this yields the central mechanistic insight: exponential topological decay is converted into graceful functional decay by threshold amplification in convergent spiking dynamics. This conversion has a simple mechanism. Let N afferents converge on a command neuron and let each synapse survive independently with probability p: expected converged drive scales linearly (Np), whereas k-hop route mass decays geometrically (p^k; 6.25% at k=4, p=0.5). The threshold maps converged drive onto output through a saturating nonlinearity, so rate varies slowly with drive near the operating point—at p=0.5 the rate retains 76.5%, more than the linear 50% drive bound. Graceful degradation therefore requires both convergent wiring, which preserves aggregate drive, and threshold dynamics, which compress drive changes into small rate changes.

V. Design Principles for Neuromorphic Embodied Controllers

Table I distills five transferable principles. P1 and P2 follow from the flat F1 and F3 curves: an evolved controller stores computation in distributed convergence rather than precise magnitudes, so implementations can prefer dense low-precision, partially defective fabrics—consistent with hardware observations [6], [8]. P3 shifts the failure budget: disconnection costs mostly latency, so real-time loops should specify tolerance as worst-case response time under edge loss—which the framework measures. P4 is the most actionable: the only

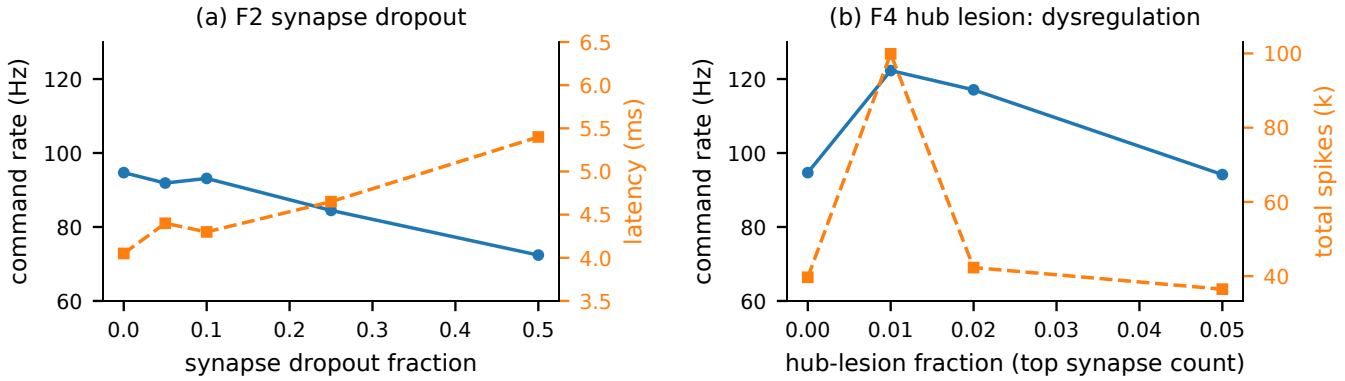


Fig. 3. Behavioral tolerance curves (medians over 12 trials; blue: rate; orange: latency or spikes). (a) Synapse dropout: 76% retained at 50% loss, latency dominant. (b) Hub lesion: dysregulation, not loss of signal.

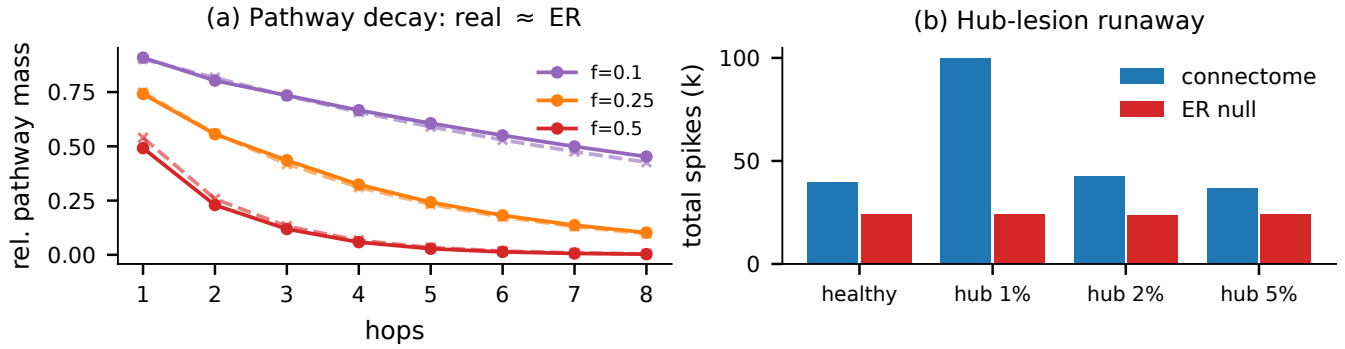


Fig. 4. Topology contrast. (a) Route mass under edge loss, normalized per graph (solid: connectome, dashed: ER): relative decay nearly identical, absolute mass 5.8–17.7× higher. (b) Activity under hub lesions: runaway only in biological wiring; ER null transmits nothing (0.0 Hz).

TABLE I
DESIGN PRINCIPLES DISTILLED FROM CONNECTOME FAULT INJECTION, WITH QUANTITATIVE ANCHORS.

#	Principle	Anchor (this study)
P1	Buy robustness with redundancy of wiring, not precision of weights	σ=0.5 lognormal noise: 100% retention
P2	Distribute command integration; no 10% slice may be load-bearing	10% random unit loss: 100% retention
P3	Budget latency, not accuracy, for disconnection faults	50% synapse loss: 76% rate, +33% latency
P4	Regulate high-fanout units; monitor envelope, not liveness	top-1% hub loss: 2.5× activity runaway
P5	Duplicate the sensory interface before the network	single-channel loss 68–79% median (weakest trial 63.1 Hz); both channels 0%

amplifying failure mode is triggered by removing the highest-fanout units, so designs should cap the influence of any single high-degree component and monitor output-envelope violation rather than component liveness—the fault presents as overactivity while all units remain alive. P5 locates the single point of failure at the interface: each sensory channel alone suffices (68–79% retention), so duplicating or cross-timing the sensory front-end buys more safety than any internal redundancy.

The study also carries a methodological message: identical relative pathway decay in both graphs, yet only the biological one transmitting a decodable command, warns against certifying embodied controllers by topology alone; and the hub-loss runaway is invisible to per-unit fault metrics, arguing for fault-injection campaigns as a standard pre-deployment stage [28], [29]. Seed-exact and seconds per trial, the sweep re-runs in minutes on any hardware mapping [5].

VI. Limitations and Conclusion

Our measurements inherit the LIF abstraction—single-compartment neurons, delta synapses, no plasticity, Poisson sensory drive; doses were bounded to keep the background regime intact, the hub analysis is descriptive, and the taxonomy does not model correlated damage or plastic recovery. In sum: an evolved 138k-neuron controller is indifferent to weight noise, distributed unit death, and even half its synapses being gone—provided convergent wiring and one sensory channel survive—and its dangerous failure is hub-loss dysregulation, not signal loss: robustness is a property of the wiring, not of precision parts.

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